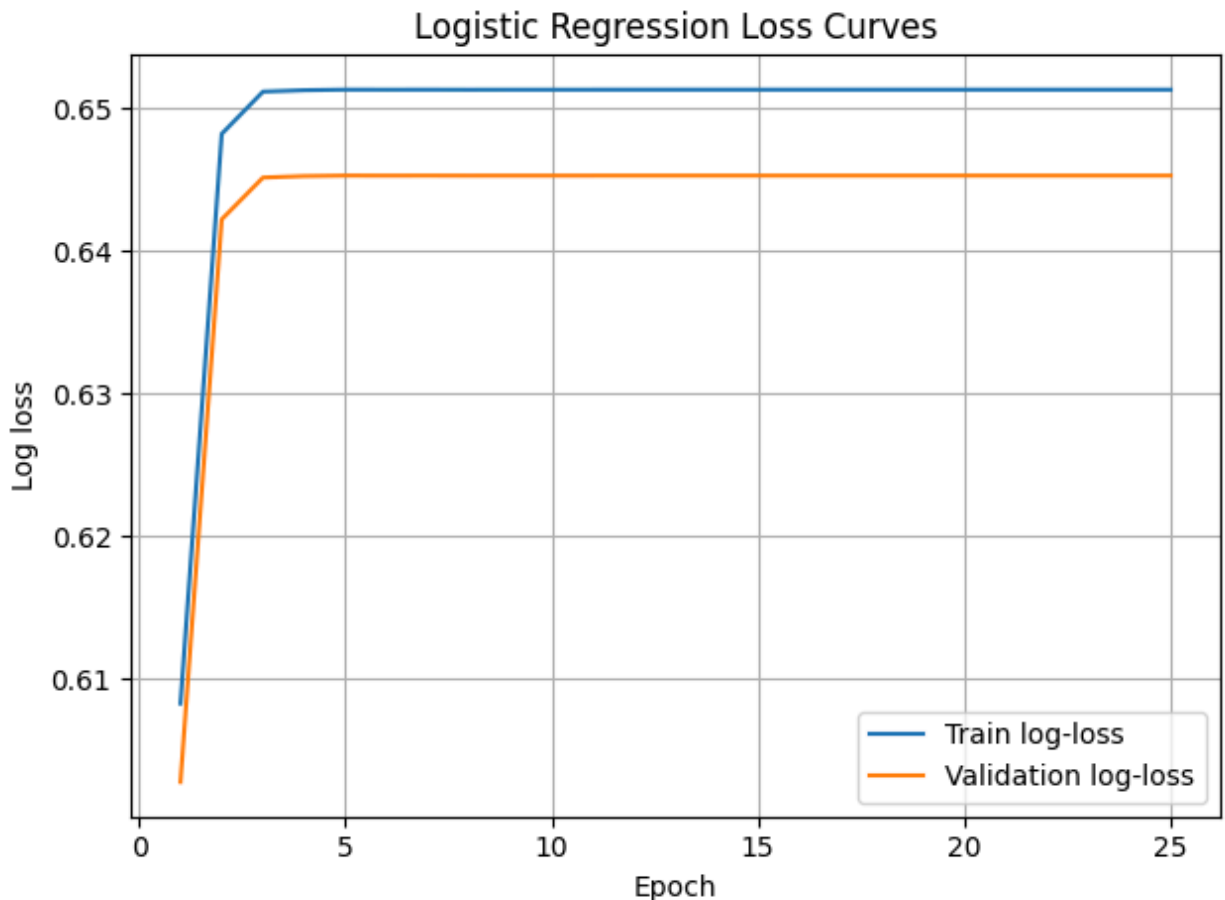


Project 1: Lung Cancer Classification Report

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2.1 Ablation study



We see here the training and validation loss curves for our logistic regression model. This plot shows that the number of epochs is pretty independent of model performance, with the model converging to a stable solution very quickly.

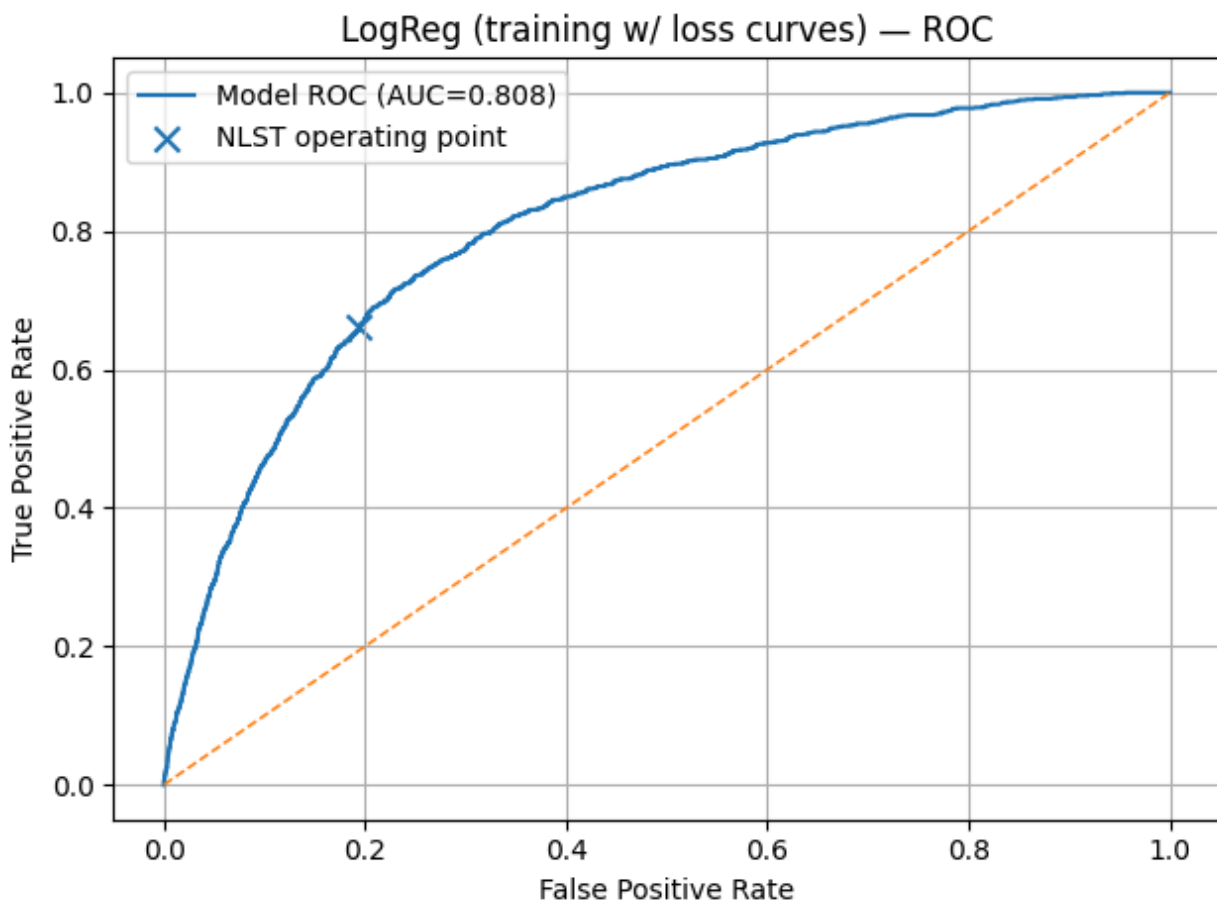
To implement this model, the most important consideration is the features of the model. We used the following features: age, number of packs smoked per day * years smoked, sex, race, current smoking status, xray_history, years of education, family history of lung cancer, and whether they currently qualify for a screening. The numerical data was normalized, and the

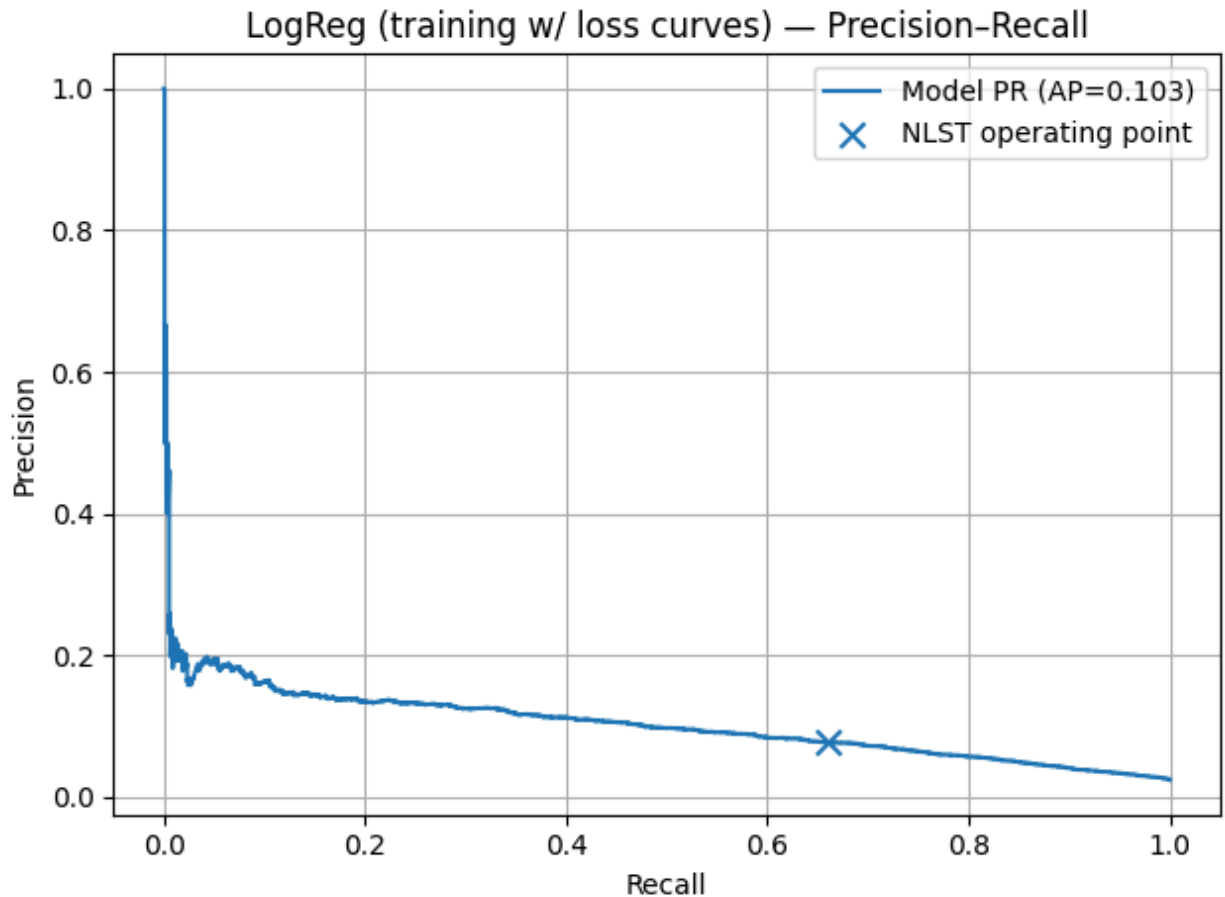
categorical data was one hot encoded. For the features with missing data, we applied imputation based on the distribution of the data that was found, assuming that someone not filling out the data was the result of randomization, and so randomly imputing the data according to the distribution from people who did submit data would reflect the actual data. From there, we performed a grid search to find an optimal regularization strength of around $\lambda = 66.667$ ($C = 0.015$ in sklearn's logistic regression) with LASSO regularization. This means this model's solution is very sparse, with model weights pushed toward 0 for weak features. The combination of LASSO and a regularization of 66.667 mean the model is unlikely to be overfit, and may have had some risk of underfitting.

The most important design decisions were what features to use, and how to handle missing data. The feature with the most influence was likely age, followed by number of packs smoked per day * years smoked.

2.2 Analyzing overall model performance

Required Plots



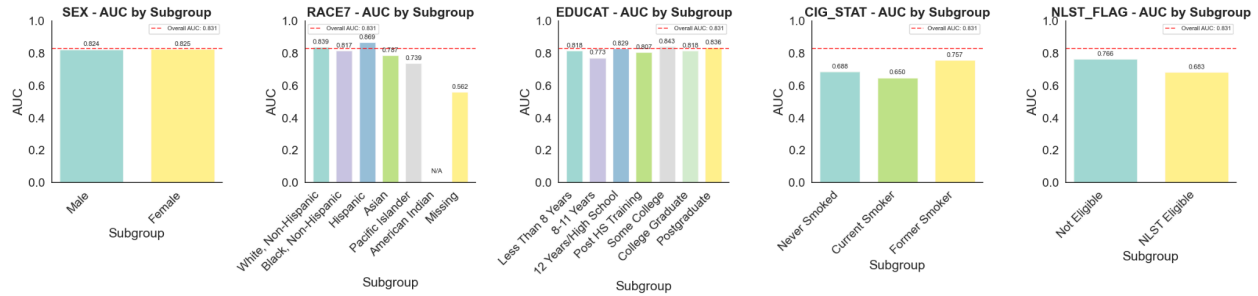


Subgroup Performance

Even though the model is performing good in general, it is important to consider its fairness among different subgroups before it's deployed to the real world. Here we consider 5 different kinds of subgroups and compared their AUC:

- Sex (sex column)
- Race (race7 column)
- Educational status (educat column)
- Cigarette smoking status (cig_stat column)
- NLST eligibility (nlst_flag column)

We do this by partitioning X_{test} into different parts according to these subgroup columns and compute AUC again individually. Here is a summary plot:



One limitation to note here is that there is missingness in the subgroup-indicating columns, which means that original performance may not match averaged subgroup performance, and some subgroups may have too few positive labels for us to be confident. Nevertheless, we still observed that there is notable performance difference between `cig_stat` and `nlst_flag` subgroup performances (auc difference > 0.05).

We think both are caused by the fact that our model are using features related to smoking status such as “`pack_years`”. This makes the model to perform better on subgroups that have smoked (also related to NLST eligibility), where it may be easier to predict since smoking is a large contributing factor to lung cancer.

2.3 Model interpretation

When testing the model on individual inputs, we found that `pack_years` had the greatest impact, achieving a test AUC of 0.7896 when used alone. `Age` was also highly influential, raising the AUC to 0.8061, followed closely by `sex`.

We ranked feature importance using two complementary approaches: ablation (as described in Section 2.1) and odds ratios derived from the model coefficients. Through ablation, `pack_years` and `age` clearly stood out, each producing AUC values above 0.6 when used individually, while most other features contributed little (remaining near a coin-toss baseline of 0.5 AUC). Odds ratios reinforced these findings: `age` (1.07) and `pack_years` (1.29) showed strong positive associations, whereas the one-hot encoded values of `sex` (0.40 and 0.38) indicated a moderate contribution.

2.4: Simulating Clinical Utility

NLST Baseline

Using the confusion matrix from sklearn, we computed the true positive (tp), false positive (fp), true negative (tn), and false negative values (fn) of the NLST criteria on the test set.

Sensitivity ($tp / (tp + fn)$): **0.66**

Specificity ($tn / (tn + fp)$): **0.81**

PPV ($tp / (tp + fp)$): **0.08**

Matched Performance

After testing our model's scores on a series of thresholds, ranging from 0 to 1 split into a series of 100 intervals, we found that within a **threshold of 0.74**, our **specificity was 0.79** and along with achieving a **sensitivity of 0.68**. As a result, we were able to match NLST's specificity within this threshold.

Risk Threshold Selection

We choose our threshold to be 0.5. This threshold yields the following precision and recall:

	precision	recall	f1-score	support
0	0.99	0.76	0.86	45350
1	0.07	0.78	0.13	1117

Even though precision is quite low (only 7% of the people invited to screen are expected to have cancer), recall is high (78% of the lung cancer will be captured by screening). The tradeoff to consider here is the cost of lung cancer screening and the danger of missing lung cancer discovery. Since the cost of the latter is inestimably high, it is reasonable to sacrifice precision to ensure more people in danger are screened.

Subgroup Performance

Next, we analyze the effect of choosing this threshold in the aforementioned subgroup.

